

# Week 15: RESTful API Design

## **1. REST Principles**

REST (Representational State Transfer) is a standard architectural style for building APIs. It uses HTTP methods and resource-based URLs to communicate between client and server. REST principles improve scalability, consistency, and simplicity in API development.

## **2. Building CRUD API (Without Database)**

A basic CRUD API was built to perform Create, Read, Update, and Delete operations without using a database. Data was stored temporarily in memory or arrays. This helped understand how API requests and responses work before integrating databases.

## **3. Status Codes, Response Formats**

HTTP status codes such as 200, 201, 400, 404, and 500 are used to indicate request results. Response formats like JSON provide structured data exchange between frontend and backend. Proper use of status codes and response formatting improves debugging and API reliability.

## **4. API Documentation Basics**

API documentation explains available routes, request methods, parameters, and expected responses. Proper documentation helps developers understand and use APIs easily. This is important for teamwork and long-term project maintenance.

## **Conclusion**

This week focused on RESTful API design. By learning REST principles, CRUD operations, status codes, response formatting, and documentation, strong API development skills were improved.